

### **Paper Questions:**

- 1- Train the LSTM to complete sentences by predicting the next word based on the context
- 2- Calculate and present the correlation coefficient between two variables in a dataset.
- 3- Describe a scenario where you'd use Azure's Cognitive Services for language analysis and explain your choice of service.
- 4- Plan and describe a solution using Azure's Anomaly Detector service to identify unusual patterns in a time-series dataset.
- 5- Develop an LSTM-based model to generate creative text passages, starting from a given seed.
- 6- Train the neural network using the MNIST dataset and observe its accuracy.
- 7- Utilize Python's data visualization libraries to create a histogram illustrating the distribution of exam scores.
- 8- Implement a basic RNN using Python and TensorFlow to predict the next word in a sentence.
- 9- Modify the RNN to perform sentiment analysis on short text samples and report the sentiment of each one.
- 10- Research and decide on the appropriate Microsoft Azure AI service for a vision-related problem, explaining your choice.
- 11- Describe how you'd leverage Azure Metrics Advisor to automatically detect and diagnose issues in business metrics.
- 12- Build a simple neural network using a deep learning framework like TensorFlow or PyTorch to classify handwritten digits.